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BENNETT UNIVERSITY THE TIMES GROUP

NAAC GRADE A+ ACCREDITED UNIVERSITY

End Semester Examinations, May 2026

Even Semester 2025-26

B.Tech.(CSE)/B.Tech.(CSE Global) – 2nd Semester

(2025CSET153) DISCRETE MATHEMATICAL STRUCTURES

Time: 2hrs. Max Marks: 40

Note: ***Student must attempt 5 Questions in total.*** *First question is compulsory and student must attempt question(s) from each section. Marks are indicated against each question.*

PART-A

Q.1 *First question will consist of 8 small questions covering entire syllabus out of which student has to attempt 6 questions.* ***2*6=12 Marks***

(a) Show that $[\neg q \wedge (p \to q)] \to \neg p$ is a tautology. [CO-1] [L-4] **2**

(b) Express the following statement using quantifier(s). Then form the negation of the quantified statement, so that no negation is to the left of the quantifier. Take the domain as all students in the college. [CO-1] [L-2] **2**

"Every student in the college has a bicycle or has a friend who has a bicycle."

(c) A relation R on $\mathbb{Z}$ is defined as aRb if and only if $a^2 - b^2$ is divisible by 4. Show that R is both reflexive and transitive. [CO-1] [L-4] **2**

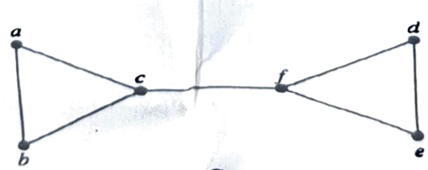
(d) Find the total number of reflexive relations on a set with n elements. [CO-1] [L-3] **2**

(e) In how many different ways can eight identical badges be distributed among three distinct volunteers if each volunteer receives at least two badges and no more than four badges. [CO-3] [L-4] **2**

(f) In a school, 300 students use WhatsApp, 250 students use Instagram, 200 students use Facebook, 80 students use WhatsApp and Facebook, 60 students use Instagram and Facebook, but no one uses WhatsApp and Instagram concurrently. Find the total number of students who uses only Facebook. [CO-3] [L-3] **2**

(g) In a gaming tournament, every player competes against every other player exactly once. A total of 45 matches are played. How many players are in the tournament? [CO-2] [L-3] **2**

(h) Give an argument to show that there is no Hamiltonian circuit in graph G. [CO-2] [L-3] **2**



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PART-B

There will be three questions each covering one specific COs and BL out of which student must attempt any two questions. Questions carrying 6 marks can be further sub divided as per the paper setter choice. ***6*2=12 Marks***

[CO-3] [L-4] **6**

Q.2 (a) Using set identities, show that $\overline{A \cup (B \cap C)} = (\bar{B} \cap \bar{A}) \cup (\bar{C} \cap \bar{A})$ for all sets A, B, C. **2**

(b) Show that the hypotheses, "If I follow my routine, I will run or do yoga. "If it is raining, I will not run". "If I am injured, I will not do yoga". "Today I follow my routine and it is raining", leads to the conclusion that, "I am not injured". Clearly mention the rule of inferences used to obtain the conclusion. 4

[CO-3] [L-3] 6

Q.3 (a) Show that if $a \equiv b(mod\ m)$ and $c \equiv d(mod\ m)$ then $ac \equiv bd(mod\ m)$. 2

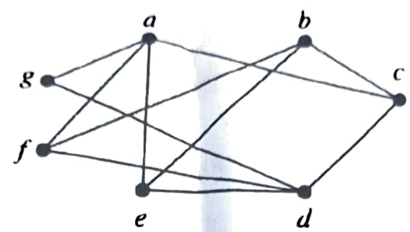
(b) Solve the system of congruences $3x \equiv 9(\text{mod } 15)$, $x \equiv 5(\text{mod } 8)$, $2x \equiv 8(\text{mod } 14)$ using Chinese Remainder Theorem. 4

Q.4 For the below graph H [CO-3] [L-3] 6

(a) construct the adjacency matrix. 1

(b) Find chromatic number. Also, list the color of each vertex. 2

(c) Define bipartite graph. Is the graph H bipartite? Justify your answer. 3



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PART-C

There are two questions with an internal choice in each of the questions covering the same (set) of COs and BL. Questions carrying 8 marks can be further sub divided as per the paper setter choice.
8*2=16 Marks

Q.5 [CO-3] [L-3] 8

(a) Show that the algebraic structures $(\mathbb{Z},*)$ forms an Abelian group, where the operation * is defined as $x * y = x + y + 1$ for any $x, y \in \mathbb{Z}$. 5

(b) Show that every cyclic group is commutative. 3

OR

(a) Let $\mathbb{N}$ be the set of all natural numbers. Then determine whether the operation * defined as $a * b = a + 2b$, for any $a, b \in \mathbb{N}$, is an associative operation. 2

(b) Show that $(Z_6, +, \times)$ forms a ~~field~~ ring under addition and multiplication modulo 6, by explicitly verifying all the ring axioms. 6

Q.6 For the partial ordering {(a, b) | a divides b} on the set A = {3, 5, 9, 15, 24, 45} [CO-2] [L-3] 8

(a) Draw Hasse diagram. 2

(b) Find maximal, minimal, greatest and least elements, if they exists. 2

(c) Find lower bounds, greatest lower bound, upper bounds, and least upper bound of the subset {3, 9, 15}, if exists. 2

(d) Determine whether the above poset is a lattice. Justify your answer. 2

OR

Given the set A = {5, 6, 7}. Consider the poset $(A \times A, \preccurlyeq)$ where the relation $\preccurlyeq$ is defined as $(a, b) \preccurlyeq (c, d)$ if and only if $a \leq c$ and $b \leq d$. [CO-2] [L-3] 8

(a) Draw the Hasse diagram of poset. 3

(b) Find lower bounds, greatest lower bound, upper bounds, and least upper bound for the subset {(5,6), (5,7)}, if exists. 3

(c) Determine whether the above poset is a lattice. Justify your answer. 2

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